

► Terminal

A text input and output environment

Features:

- (i) several libraries
- (ii) programming via Git
- (iii) speed
- (iv) Access

• Different terms:

1. Command line:

any interface that is used by entering textual commands (gen. windows centric)

2. Terminal: This is a type of command line (gen. Mac centric)

3. Console: A command line interface used to work with your computer

4. Shell: A program running on terminal

5. Bash: A popular shell on Mac OS/Unix

6. Z-shell: Another shell

mkdir command

► Commands

1. Navigation

cd = change directory

cd Desktop

cd .. = back button

• To check the text → pwd

2. ls = list files (show my files)

3. pwd = print working directory (where am I?)

4. clear = clear screen

5. Paths

Absolute & Relative

► Paths In Navigation

cd Desktop / Delta (relative path)

cd /Users/shradhakapra/Desktop (absolute path)

/ = root directory

~ = home directory

► Making Directories

mkdir

► flags

Flags are characters that we pass with commands to modify their behaviours

manual command

man ls - give info about ls command

man mkdir - give info about

ls -l → list in long format

ls -a

ls -la

with flags

include dir. entries
whose names begin
with a (.)

Name: ls -l

(list directory contents) hidden file

Combine Command

ls -la

► touch command

used to create files

touch index.html

change file access and modification times. If any file doesn't exist, it is created with default permission

► Deleting Files and folders

rm → remove files

rmdir → remove empty folders

rm -rf → remove any folders

recursive delete
forced

► clone command

Go to code; copy https path
url

In terminal,

git clone "url"

► What is Git?

Free & open source
version control system

tools that helps to track
changes in code

• NOTES

(i) track history

(ii) help to collaborate

(iii) fast and scalable

► What is Github?

website where we host
repositories online

► configuring Git

git config --global user.name
"MyName"

git config --global user.email
"someone@email.com"

► Basic Commands

• clone - cloning a
repository on our local
machine; git clone "url"

• status - displays that
the state of the code

→ git status

untracked

↓
unmodified

↓
modified

↓
staged

add the
file or
remove

add the
file

stage
the file

Commit

► **add** - adds new or changed files in your working directory to the Git staging area

- `git add <file name>`

► **commit** - it is the record of change

- `git commit -m "some message"`

► **push** - upload local repo content to remote repo

- `git push origin main`

► **init command** - initialise repo in git

- `git init`

- `git remote add origin <link>`

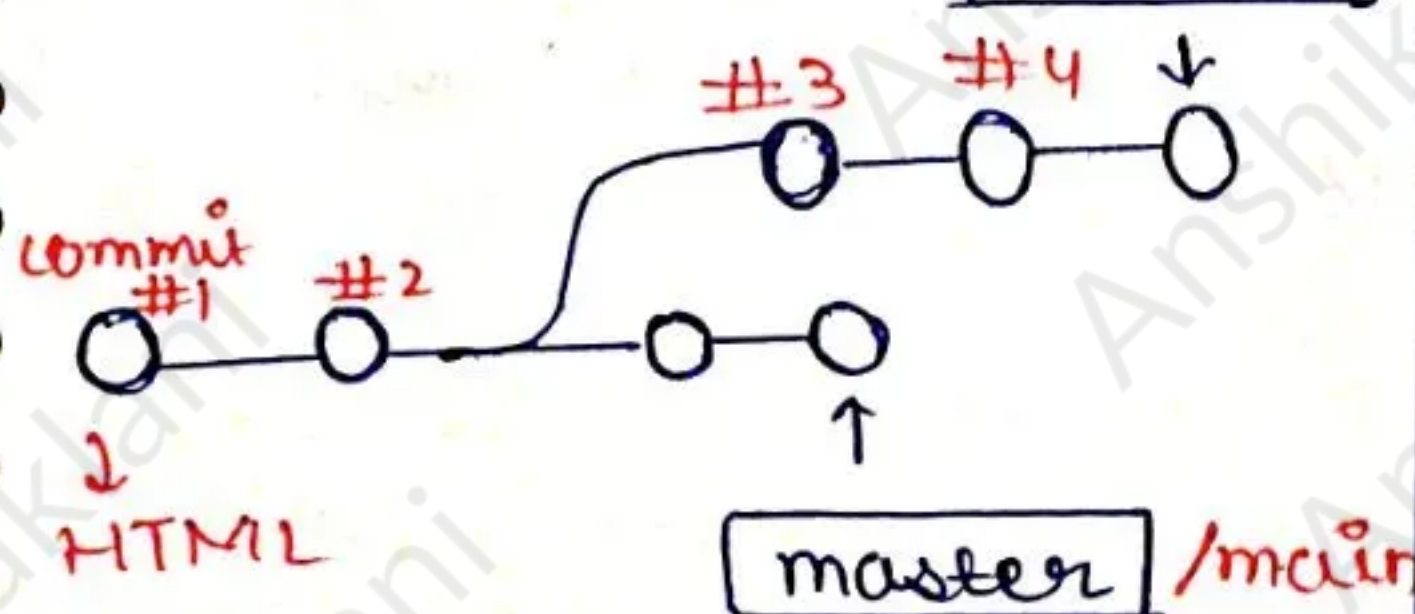
- `git remote -v` (to verify)

- `git branch` (to check branch) ^{remote}

- `git branch -M main` (to rename branch)

- `git push origin main`

► **Git Branches**



NOTE: if you want that do not want to write "git push origin main" again and again SET THE UPSTREAM:

1. `git push -u origin main`

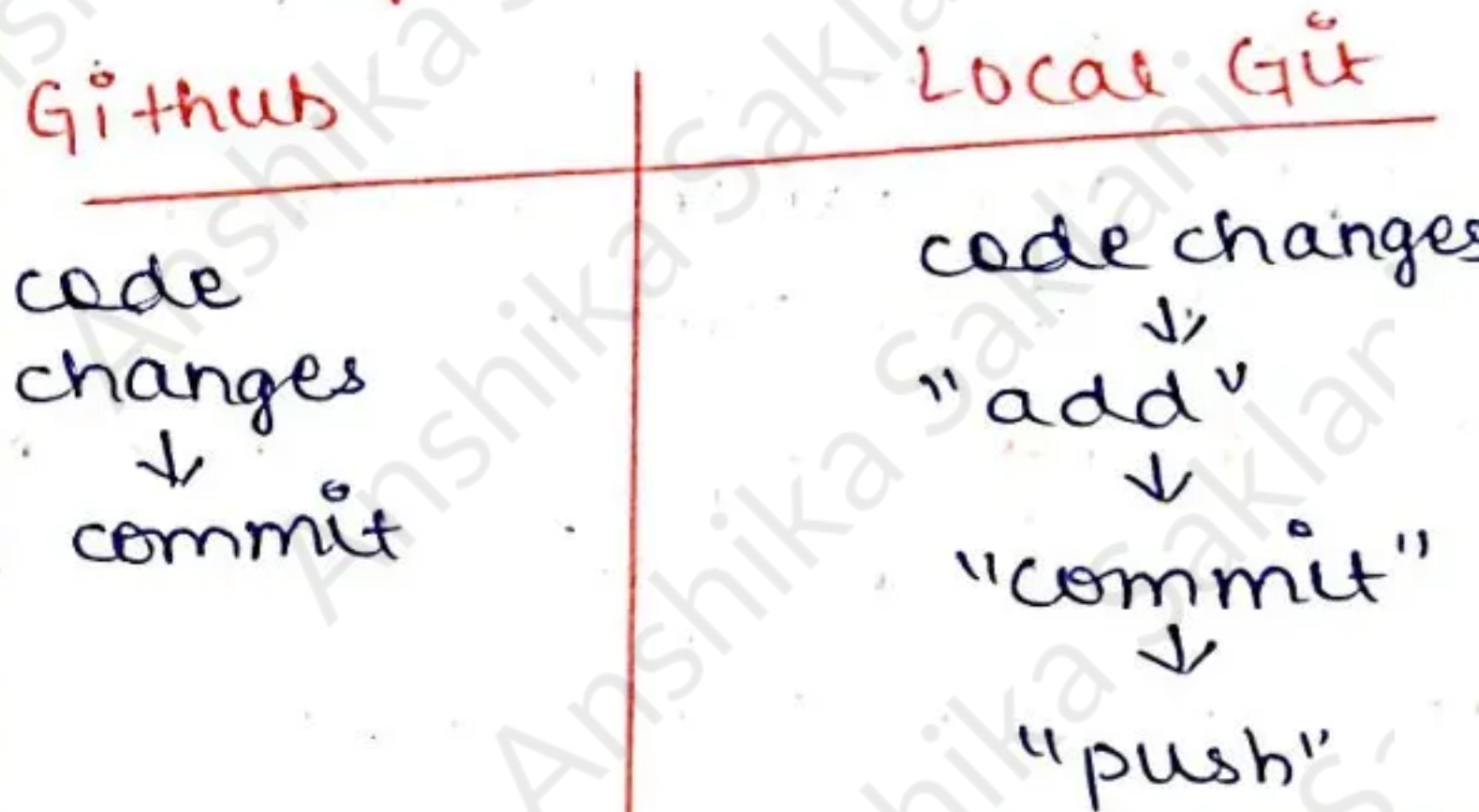
2. `git status`

3. `git commit -am "added"`

↓
add and commit both at all once

4. `git push`

► **Workflow**



► **Branch commands**

`git branch` (to check branch)

`git branch -M main` (to rename branch)

`git checkout <branch name>` (to navigate)

`git checkout -b <new branch name>` (to create new branch)

`git branch -d <branch name>` (to delete branch)

► Merging Branches / code

git diff <- branch name -> (to compare commits, branches files and more)

git merge <- branch -> (to merge 2 branches)

OR

Create a PR

► Pull request : It lets you tell others about changes you've pushed to a branch in a repository on Github

Basic Commands : git pull origin main



used to fetch and download content from a remote repo and immediately update the local repo to match that content.

► Merging conflicts : an event that takes place when Git is unable to automatically resolve differences in code between two commits.

git merge <- branch name ->

► Case 1 : staged changes

git reset <- file name ->
git reset

► Case 2 : committed changes (for one commit)

git reset HEAD~1 ^{ptr (last commit)}

► Case 3 : committed changes (for many commits)

git reset <- commit hash ->

git reset --hard
<- commit hash ->

NOTE : git log



gives step by step commit till now!

What is Forking?

A fork is a new repository that shares code and visibility settings with the original "upstream" repository.

- Fork is a rough copy.